



L20: Neon City Graffiti Tool

Final Graphics
Project





Project Overview

Build a digital spray paint system!

Features:

-  Draw with arrow keys
-  Change colors with number keys
-  Pen up/down control
-  Clear canvas

Combines: Objects + Methods + Events + Loops



Step 1: Setup

```
import turtle

# Create the pen object
pen = turtle.Turtle()
pen.shape('circle')
pen.color('cyan')
pen.pensize(3)
pen.speed(0)

turtle.done()
```

Foundation complete - pen ready to draw!



Step 2: Color Palette

```
def color_cyan():  
    pen.color('cyan')  
  
def color_magenta():  
    pen.color('magenta')  
  
def color_yellow():  
    pen.color('yellow')  
  
screen = turtle.Screen()  
screen.bgcolor('black') # Neon canvas!  
screen.listen()  
screen.onkey(color_cyan, '1')  
screen.onkey(color_magenta, '2')  
screen.onkey(color_yellow, '3')
```



Step 3: Movement Controls

```
def move_forward():  
    pen.forward(20)
```

```
def move_backward():  
    pen.backward(20)
```

```
def turn_left():  
    pen.left(15)
```

```
def turn_right():  
    pen.right(15)
```

```
screen.onkey(move_forward, 'Up')  
screen.onkey(move_backward, 'Down')  
screen.onkey(turn_left, 'Left')  
screen.onkey(turn_right, 'Right')
```



Step 4: Pen Control

```
def pen_up():  
    pen.penup()    # Stop drawing  
  
def pen_down():  
    pen.pendown()  # Start drawing  
  
def clear_canvas():  
    pen.clear()    # Erase everything  
  
screen.onkey(pen_up, 'u')  
screen.onkey(pen_down, 'd')  
screen.onkey(clear_canvas, 'c')
```

Now you can move without drawing!



Complete Control Scheme

Key	Action
1, 2, 3	Change color
Arrow Keys	Move/turn pen
U	Pen up (stop drawing)
D	Pen down (start drawing)
C	Clear canvas



Pro Features

```
# Speed control
pen.speed(0) # Instant = smoother drawing

# Thicker lines
pen.pensize(5)

# Hide the turtle cursor
pen.hideturtle()

# Custom background
screen.bgcolor('#0a0a0a')
```




Art Techniques

Dotted Lines:

```
for i in range(20):  
    pen.pendown()  
    pen.forward(5)  
    pen.penup()  
    pen.forward(5)
```

Gradient Effect:

```
colors = ['cyan', 'blue', 'magenta', 'red']  
for i in range(40):  
    pen.color(colors[i % 4])  
    pen.forward(10)
```

✓ Project Complete!

You Built:

- ✓ Interactive paint program
- ✓ Color switching system
- ✓ Movement controls
- ✓ Pen state management
- ✓ Clear function

This is a REAL application! Share your art!



Extension Challenges

1. Add more colors (4-9 keys)
2. Create shape stamps (press 's' for star)
3. Save/load drawings (advanced!)
4. Fill tool (`begin_fill()` , `end_fill()`)

Get creative - it's YOUR tool!